LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	16
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Wheels, types of wheels and description of typical main wheel
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Wheels, types of wheels and description of typical main wheel
5. Specific Objectives	17.1.3.1 Trainee will be able to describe typical alighting gears of an aircraft and its construction.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Examination of typical main wheel assembly components; Safety: Wear eye protection; Instructor supervision required
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of aircraft landing gear?
(b) Name two types of wheel assemblies used in aviation.
(c) Why are wheels critical for aircraft ground operations?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Classification of aircraft wheels • Construction of typical main wheel • Brake assembly integration • Wheel hub and axle details • Landing gear shock absorption	L & Ds	PPT, IB	15 min
(b)	• Detailed analysis of wheel rim types • Function of wheel bearings • Inspection criteria for wheel components • Maintenance requirements for brakes • Operational limits and load ratings	L & Ds	PPT, PDF	20 min
(c)	• Identification of wheel wear patterns • Common defects in main wheel assemblies • Procedures for wheel removal and installation • Torque specifications for wheel nuts • Documentation of wheel maintenance actions	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the construction sequence of a typical main wheel.
(b) Describe the role of wheel bearings in aircraft operation.
(c) List three common defects found during wheel inspections.

12. Home Assignment and Follow Up	Prepare a brief report on tyre pressure variation with all-up weight as covered in the next lesson plan (LP 17).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Tyre and tubes Tyre variation of Tyre pressure with all up weight, creep and creep marking on wheels	Refer to the master syllabus reference list	LP 17

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min